

SHRESTH SINGH

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EDUCATION

Duke University — Durham, NC | B.S. Mechanical Engineering

Expected May 2029

GPA: 3.8 / 4.0 · ACT: 36

Relevant Coursework: MAT-218 (Linear Algebra), EGR-101/EGR-121 (Engineering Design Courses), GER-210 (German Language)

RESEARCH

Undergraduate Researcher — Satellite Imaging, Critical Materials Hub (Data+)

Summer 2026

- Built a Python geospatial pipeline (geopandas, rasterio, py3drep, SciPy) to estimate recoverable critical-mineral volumes across ~100 abandoned mine-waste sites in AZ, NC, VA, and MT, benchmarked against USGS field-derived volumes.
- Reconstructed buried pre-mining terrain from digital elevation models via thin-plate-spline interpolation; validated reconstructions against held-out natural ground to within ~0.2 m. Documented the full pipeline across nine reproducible notebooks.
- Cut regional volume bias from roughly -58% to within a few percent of the USGS benchmark using a log-ratio correction, and characterized a stable per-site error floor across five independent tests (sample size, error model, DEM resolution, spline kernel, gating). Co-authored a research poster with faculty.

ENGINEERING TEAMS

Vehicle Dynamics Engineer — Duke Motorsports (Formula SAE)

2025 – Present

Competed at Formula SAE Michigan, May 2026

- Calculated front-upright and wheel-hub load cases for FEA: scaled prior-season g-g load data by 1.15 and ran 28 load cases through the team's suspension-force model to extract per-corner member forces feeding the SolidWorks upright analysis.
- Wrote a Python automation script (Excel COM) to batch all 28 load cases and export a clean results workbook, replacing a manual, error-prone process. Validated script outputs against the previous year's load cases.
- Authored a design report evaluating camber-shim placement across load path, serviceability, and effects on scrub radius and kingpin inclination. Revised the upright design, relocating the wheel-speed sensor to improve mechanical adjustability.

Structural & Aerospace Engineer — Duke Aero (Rocketry)

2025 – Present

- Designed and fabricated a Von Kármán carbon-fiber nose cone (CAD → male 3D-printed plug → female fiberglass mold → vacuum layup) to <1 mm tolerance; validated in ANSYS (stress, strain, deformation) to within ~2%.
- Produced forged carbon-fiber airbrake paddles via compression molding and co-led carbon body-tube layup for flight-ready airframe sections used in competition. Documented layup procedures and resin-system composition for team reference.

ADDITIONAL EXPERIENCE

Data Analyst Intern — Relanto

Jun – Dec 2024

- Ran SQL data-integrity checks across relational datasets and wrote Python scripts to test model behavior under parameter variation and surface systematic failure modes.

Founder & Lead Tutor — Apex Academix

2021 – 2024

- Founded a tutoring service for 20+ students; built scheduling and lesson systems and tracked outcomes with data, driving multi-grade improvements and 300+ SAT / 10+ ACT point gains.

SELECTED PROJECTS

Electronically Operated Launcher — EGR 101, Engineering Design & Communication

Fall 2025

- Designed and prototyped an electronically operated launcher, modeling system behavior and control logic to reach 90% empirical accuracy against predicted performance, and evaluated model residuals to account for the remaining error.
- Analyzed energy transfer and efficiency in the spring-actuated mechanism using physical and mathematical models; implemented the design in Onshape and Arduino under explicit tolerancing.

SKILLS

Engineering Process: SolidWorks, Onshape, ANSYS (Finite Element Analysis), Arduino, GD&T and tolerancing

Programming Tools: Python (NumPy, pandas, geopandas, rasterio, Matplotlib), SQL, C++ (fundamentals), MS Suite, Git, Jupyter

Fabrication: carbon-fiber layup, composite molding, 3D printing, technical documentation, safety factor validation

Languages: English (fluent), German (intermediate), Hindi (conversational)

LEADERSHIP & ACTIVITIES

Martial Arts Instructor, Vision Martial Arts (40+ students, ages 3–75) · Lead Volunteer, DOCNC (\$5,000+ raised) · Alpha Phi Omega service · DECA & FBLA state-level awards · Duke Cyber